

```
1  /*
2   * Name : Nachiket Singla
3   *Roll no: 1024170077
4   */
5  public class LeapYear{
6      Run | Debug
7      public static void main(String[] args) {
8          int year=2024;
9          boolean isLeap=false;
10         if(year%4==0)
11         {
12             if(year%100==0)
13             {
14                 isLeap= (year%400==0);
15             }
16             else{
17                 isLeap=true;
18             }
19         }
20         if(isLeap)
21             System.out.println(year + "is a leap year.");
22         else
23             System.out.println(year +"is not a leap year.");
24     }
```

```
/*
```

```
* Name: Nachiket Singla
```

```
💡 Roll No: 1024170077
```

```
*/public class AlphabetCheck {
```

Run | Debug

```
public static void main(String[] args) {
```

```
    char c='*';
```

```
    if((c>='a' && c<='z') || (c>='A' && c<='Z')){
```

```
        System.out.println(c+" is an alphabet.");
```

```
    }
```

```
    else
```

```
    {
```

```
        System.out.println(c+" is not an alphabet");
```

```
    }
```

```
}
```

```
}
```

```
1  /*
2  * Name: Nachiket Singla
3  * Roll No: 1024170077
4  */ public class Factorial {
```

Run | Debug

```
5  public static void main(String[] args) {
6      int num = 5;
7      long factorial =1;
8      for(int i=1 ;i<=num;++i)
9      {
10         factorial *=i;
11     }
12     System.out.println("Factorial of "+num+" = "+factorial);
13 }
14
15 }
16
```

```
/*  
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 * Roll No: 1024170077  
 */
```

```
public class Fibonacci {
```

Run | Debug

```
public static void main(String[] args)  
{  
    int n=10,secondTerm=1,firstTerm=0;  
    System.out.println("Fibonacci Series till "+n+"terms:");  
    for(int i=0;i<=n;i++)  
    {  
        System.out.println(firstTerm+",");  
  
        int nextTerm=firstTerm+secondTerm;  
        firstTerm=secondTerm;  
        secondTerm=nextTerm;  
    }  
}
```

```
/*  
 * Name: Nachiket Singla  
 * Roll No: 1024170077  
 */
```

```
public class GCD {
```

Run | Debug

```
public static void main(String[] args)  
{  
    int n1=81, n2=153;  
    //Euclidean algorithm  
    while(n1!=n2)  
    {  
        if(n1>n2)  
            n1=n1-n2;  
        else  
            n2=n2-n1;  
    }  
    /*  
    while(n2!=0)  
    {int temp=n2;  
    n2=n1%n2;  
    n1=temp; //n1 is GCD now } */  
    System.out.println("GCD is:"+n2);  
}
```

```
}
```

```
/*
```

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```

```
*/
```

```
public class LCM {
```

Run | Debug

```
public static void main(String[] args) {
```

```
    int n1=72,n2=120,gcd=1;
```

```
    //first find gcd
```

```
    for(int i=1;i<=n1 && i<=n2;++i)
```

```
    {
```

```
        if(n1%i==0 && n2%i==0)
```

```
        {
```

```
            gcd=i;
```

```
        }
```

```
    }
```

```
    int lcm =(n1*n2)/gcd;
```

```
    System.out.printf(format: "The LCM of %d and %d is %d",n1,n2,lcm);
```

```
}
```

```
}
```

```
/*
```

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```

```
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```

```
*/public class CountDigits {
```

Run | Debug

```
public static void main(String[] args) {
```

```
    int num=345289;
```

```
    int count=0;
```

```
    while(num!=0)
```

```
    {
```

```
        num/=10;
```

```
        ++count;
```

```
    }
```

```
    System.out.println("Number of digits:"+count);
```

```
}
```

```
}
```

```
/*
```

```
* Name: Nachiket Singla
```

```
* Roll No: 1024170077
```

```
*/public class Reverse {
```

Run | Debug

```
public static void main(String[] args) {
```

```
    int num=1234567,reversed=0;
```

```
    while(num!=0)
```

```
    {
```

```
        int digit=num%10;
```

```
        reversed=reversed*10+digit;
```

```
        num/=10;
```

```
    }
```

```
    System.out.println("Reversed Number:"+reversed);
```

```
}
```

```
}
```

```
/*
```

```
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```

```
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```

```
*/public class PowerCalculation {
```

Run | Debug

```
public static void main(String[] args) {
```

```
    int base=3,exponent=4;
```

```
    long result=1;
```

```
    while(exponent!=0)
```

```
    {
```

```
        result *=base;
```

```
        --exponent;
```

```
    }
```

```
    System.out.println("Answer =" +result);
```

```
}
```

```
}
```

```
/*
💡 Name: Nachiket Singla
* Roll No: 1024170077
*/

public class Palindrom {
    int reversedNum = Palindromecheck.main(String[])

    Run | Debug
    public static void main(String[] args) {
        Source: enterprise web applicaton_ded17103

        int num=121, reversedNum=0, remainder;
        int originalNum=num;
        while ((num!=0)) {
            remainder=num%10;
            reversedNum=reversedNum*10+remainder;
            num/=10;
        }
        if(originalNum==reversedNum)
        {
            System.out.println(originalNum+" is a palindrome");
        }
        else{
            System.out.println(originalNum+" is not a Palindrome");
        }
    }
}
```